

HOMEWORK 1

1. Prove that an Artinian integral domain is a field.
2. Let $f : M \rightarrow M$ be a surjective endomorphism of a Noetherian module M over a commutative ring. Show that f is an isomorphism.
3. Let M be a Noetherian module over a commutative ring R . Show that for every multiplicative subset $S \subset R$, the $S^{-1}R$ -module $S^{-1}M$ is Noetherian.
4. Show that the ring of all 2×2 matrices $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ such that $a \in \mathbb{Z}$ and $b, c \in \mathbb{Q}$ is right Noetherian but not left Noetherian.
5. Show that the ring of all 2×2 matrices $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ such that $a \in \mathbb{Q}$ and $b, c \in \mathbb{R}$ is right Artinian but not left Artinian.
6. Let F be an algebraically closed field and let $M \subset F[x_1, x_2, \dots, x_n]$ be a maximal ideal. Prove that there exist $a_1, a_2, \dots, a_n \in F$ such that $M = \{f \in F[x_1, x_2, \dots, x_n] : f(a_1, \dots, a_n) = 0\}$.
7. Let F be a field and let $M \subset F[x_1, x_2, \dots, x_n]$ be a maximal ideal. Prove that the quotient ring $F[x_1, x_2, \dots, x_n]/M$ is a finite field extension of F .
8. Let I be an ideal of a commutative ring R . Set

$$\sqrt{I} = \{a \in R \text{ such that } a^n \in I \text{ for some } n \in \mathbb{N}\}.$$

Prove that $\sqrt{I}$ is an ideal in R (called the *radical of I*).

9. Let F be an algebraically closed field, $I \subset F[x_1, x_2, \dots, x_n]$ an ideal. Denote by $S(I)$ the subset in F^n consisting of all n -tuples $(a_1, a_2, \dots, a_n) \in F^n$ such that $f(a_1, a_2, \dots, a_n) = 0$ for all $f \in I$. A subset $S \subset F^n$ is called *closed* if $S = S(I)$ for some ideal $I \subset F[x_1, x_2, \dots, x_n]$. Prove that the union of two closed subsets and the intersection of any family of closed subsets is also closed.
10. For any closed subset $S \subset F^n$ denote by $I(S)$ the set of all polynomials $f \in F[x_1, x_2, \dots, x_n]$ such that $f(a_1, a_2, \dots, a_n) = 0$ for all $(a_1, a_2, \dots, a_n) \in S$. Show that $I(S)$ is an ideal in $F[x_1, x_2, \dots, x_n]$ such that $I(S) = \sqrt{I(S)}$ (i.e. $I(S)$ is a *radical* ideal).